



Science of Corporate Technical Documentation for Software Engineers -Practices-

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For Japanese version, click [here](#)

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About Me

- Accounts
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 - LinkedIn: [@hayat01sh1da](#)
 - Speaker Deck: [@hayat01sh1da](#)
 - Docswell: [@hayat01sh1da](#)
- Occupation: Software Engineer
- Things I Am Into
 - Linguistics
 - Singing at Karaoke
 - Listening to Music
 - Watching Movies
 - Cat Feeding



Licences / Certifications

- English
 - TOEIC® Listening & Reading 915: Certified on December 2019
- Engineering
 - Information Security Management: Certified on November 2017
 - Applied Information Technology Engineer: Certified on June 2017
 - Fundamental Information Technology Engineer: Certified on November 2016
 - IT Passport: Certified on April 2016
- Others
 - Abacus 2nd Class: Certified on June 2002
 - Mental Arithmetic 3rd Class: Certified on February 2001

Skill Stack

- Languages
 - Japanese: Native Proficiency
 - English: Full professional Proficiency
- Development
 - **Backend: Ruby(Ruby on Rails), Python**
 - Frontend: TypeScript + React.js, TypeScript +Vue.js
 - **Database: MySQL**, PostgreSQL, MongoDB
 - **Architecture: Monolith, Modular Monolith**
 - Hosting: AWS ESK
 - Versioning: Git, GitHub
 - **CI/CD: GitHub Actions**, ArgoCD
 - Monitoring: Datadog, Sentry

Work Histories

Period	Genre	Position	Jobs
Aug 2021 - Present	SaaS(Educational Service)	Software Engineer	• Backend Development • Frontend Development • CI/CD Maintenance • Weekly Release Manager • Troubleshooting • Documentation • Technical Debt Resolution • Tech Blog Entries • LT Speaker
Feb 2020 - Dec 2020	SaaS(Chatbot Platform)	Backend Engineer	• Backend Development • Documentation • Technical Debt Resolution • Inspection of Alternative API

Work Histories

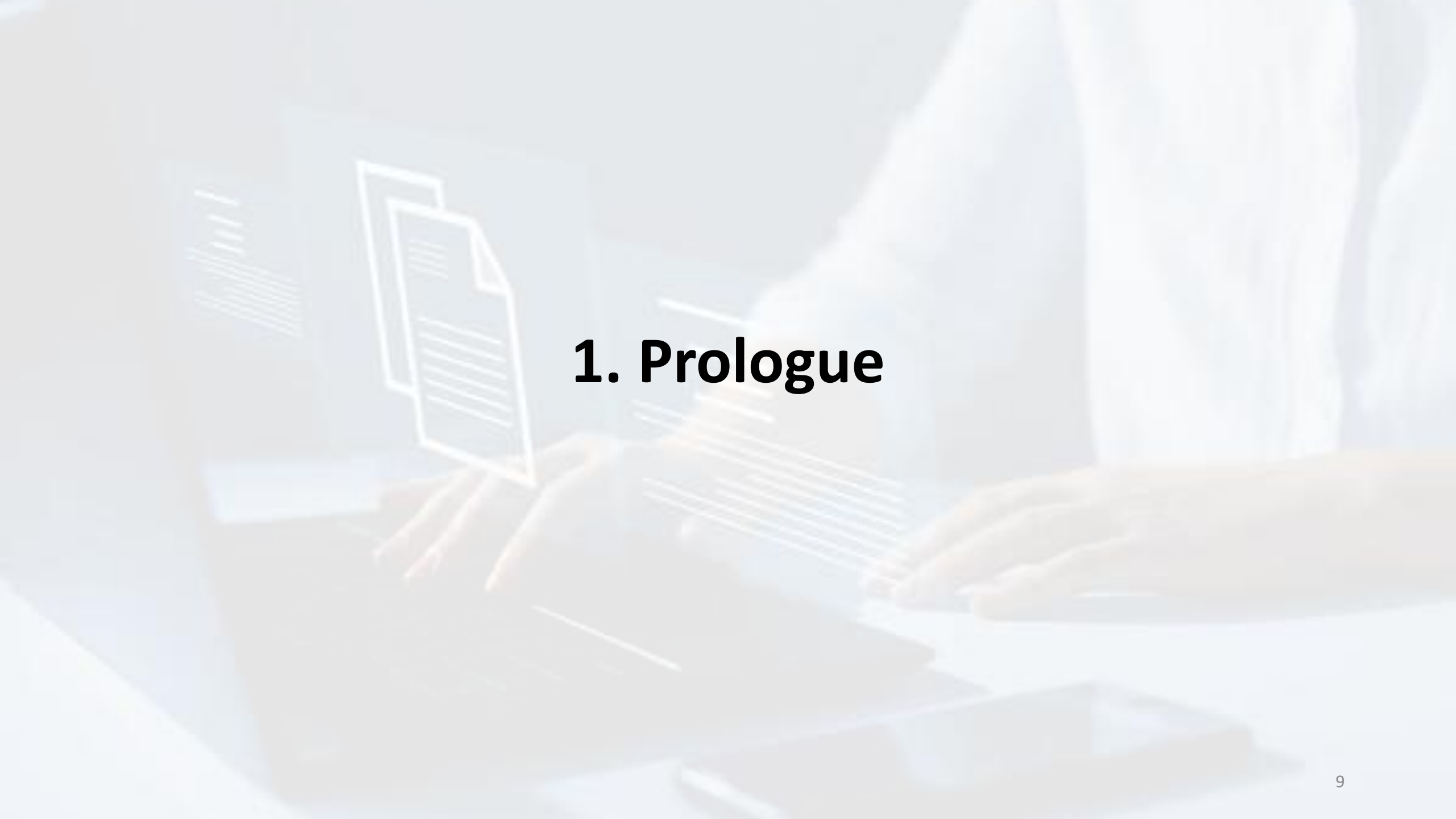
Period	Genre	Position	Jobs
Jun 2018 - Jan 2020	Contract-basis Developer	Software Engineer	• Backend Development • Frontend Development • Troubleshooting • Quality Assurance • Tech Blog Entries
Apr 2016 - Jan 2018	System Engineering Service	System Engineer	• Corporate Account Admin • Windows Server Maintenance • Security Admin Assistant • Internationalisation / Localisation

International-Exchange Activities

- Activities in University
 - English Linguistics Seminar(Focusing on Mass Media English)
 - International-Exchange Clubs(The 2nd Year)
 - [International-Exchange Programmes conducted by Japan Cabinet Office](#)(2013 - 2016)
 - Japanese Linguistics Course(The Final Year)
- Overseas Life Experience
 - Working Holiday in Australia(April 2014 - March 2015)
 - Language School for 1 month in Sydney
 - Work for 6 Months in [Hamilton Island Resort](#)
 - Volunteering for 1 Month as Assistant Teacher of Japanese Language at [St Ives High School](#) in NSW
- Other Activities
 - Keep Everyday Journal in English (April 2014 - Present)
 - [Sunrise Toastmasters Club](#)(February 2017 - March 2018)
 - [Vital Japan](#)(January 2018 - July 2019, October 2022 - February 2023)
 - Self Learning and Training of English Language
 - Participation Ruby-Related Tech Conferences

Agenda

1. [Prologue](#)
2. [Top 5 Inevitable Trade-Offs](#)
3. [Practical 10 Steps for Well-Organised Documentation](#)
4. [Summary](#)
5. [Epilogue](#)
6. [References](#)

A person wearing a white lab coat is sitting at a desk, typing on a laptop. The image is faded and has a light blue tint. Overlaid on the laptop screen is a white outline of a document with lines of text. The text '1. Prologue' is centered on the image in a bold, black font.

1. Prologue

1. Prologue

I shared the following items in the last deck [*Science of Technical Documentation for Software Engineers -Practices-*](#).

1. [Top 6 Required Factors for Technical Documentation](#)

1. [To Be Scientific](#)
2. [To Be Efficient](#)
3. [To Be Impressive](#)
4. [To Clarify Who Readers Are](#)
5. [To Communicate between Technicians and Non-Technicians](#)
6. [To Design Appropriate Workflows](#)

2. [Top 5 Required Skills for Technical Documentation](#)

1. [Ability to Get Readers Well-Informed](#)
2. [Ability to Collaborate with Stakeholders](#)
3. [Ability to Solve a Problem by Yourself](#)
4. [Ability to Manage Resources and Costs](#)
5. [Ability to Predict Potential Issues](#)

3. [Top 3 Benefits of Well-Organised Technical Documentation](#)

1. [Growth of Trust in Your Organisation](#)
2. [Mid-to-Long Term Advancement of Productivity](#)
3. [Mid-to-Long Term Cost Reduction](#)

1. Prologue

I also gave notice of [*Top 5 Inevitable Trade-Offs*](#).

This time, I analysed the trade-offs we are forced to face in carrying out the practices.

Considering it, I will discuss what are the most efficient procedures to take in order to complete documentation keeping the blockers minimum and the benefit maximum based on the following same reference as adopted in the last deck .

[*Basics Of Technical Documentation For Engineers*](#), Ayca LITTLE, Published in [*DatasheetEST by TDSmaker*](#), 25 April 2018

A person wearing a white lab coat is shown from the chest down, sitting at a desk and typing on a laptop. The image is faded and has a soft, ethereal quality. Overlaid on the left side of the image is a stylized, white-outlined icon of a folder or document with lines representing text. The background is a light, neutral color.

2. Top 5 Inevitable Trade-Offs

2. Top 5 Inevitable Trade-Offs

The top 5 inevitable trade-offs the write of a technical document are as follows.

1. Cynical Recursive Dependency of Work
2. Agreements to Make with Team Members
3. The More Stakeholders, The Less Quality
4. Responsibility of Continuous Maintenance
5. Difficulty in Find Out the Upper Limit of Investment

2. Top 5 Inevitable Trade-Offs

1. Cynical Recursive Dependency of Work

If certain knowledge or management processes are heavily dependent on a specific member, **software engineers must take action to resolve the dependency so that they enable the stakeholders to refer to or inherit it.**

Corporate technical documentation is one of the means to exterminate such a dependency; however, only the writer of a technical documentation is familiar with the target knowledge and management process until the work is done.

The writer can delegate some tasks to other members if the knowledge and management process are not complicated to deal with.

Otherwise, **the lecture costs will easily surpass those of the documentation**, so the writer will not be allowed to build a coalition.

It is the very reason this work to exterminate dependencies is recursively so dependent.

2. Top 5 Inevitable Trade-Offs

1. Cynical Recursive Dependency of Work

The best solution to the trade-off is to choose either of the following strategies.

1. Accept documentation costs and make it done by themselves.
2. Pay lecturing costs and distribute balanced loads to other collaborative writers.

I myself chose to take Strategy 1 and succeeded in acquiring not just comprehensive and deep domain knowledge but skills to organise and manage complicated and profound information in exchange for serious damage on my mental health.

If the writer does not stick to a self-contained working style, Strategy 2 will be much wiser choice in terms of their mental health.

2. Top 5 Inevitable Trade-Offs

2. Agreements to Make with Team Members

In the case of corporate technical documentation by team, **the quality of deliverables heavily depends on each member whereas the loads of the work can be equally distributed.**

Particularly, the huger the difference in documentation experiences is among the members, the more likely [the review phrase](#) is to fail to resolve ununified formats, description and expressions.

Creating a documentation guideline might be a good idea to watch grammatical and collocational mistakes just like a design guideline for User Interface.

However, **each writer, in turn, is highly likely to feel uncomfortable in the work itself under such a circumstances where they are not trusted and ruled by the strict guideline.**

2. Top 5 Inevitable Trade-Offs

2. Agreements to Make with Team Members

When I took initiative of team documentation, I made the following basic agreements with other members.

1. The Entire Structure of Sections and Chapters
2. The Working Costs in Each Section and Chapter(Story Point)
3. The Assignment Scheme and Assignee of Each Section or Chapter
4. Commonly Used Chart Drawing Tools
5. [Workflows](#)

Thanks to only the basic agreements in stead of strict rules, a comfortable working experiences as well as insurance of the quality were successfully secured.

2. Top 5 Inevitable Trade-Offs

3. The More Stakeholders, The Less Quality

A personal document can be personally updated and a team document can also be modified by the team as long as an agreement is made.

However, if members from another division or a company are also involved in a project, a large number of stakeholders makes it a lot harder to have an accurate and timely communication of the latest information because it can be passed from a member to another just like Broken Telephone.

What is worse, it is also painstaking to arrive at an agreement with changed or updated information.

That is to say, **management of shared information among such huge stakeholders is never easy to deal with.**

2. Top 5 Inevitable Trade-Offs

3. The More Stakeholders, The Less Quality

The best solution to this trade-off is to focus on the readers who really need to refer to a corporate technical document and minimise the number of the stakeholders.

Concretely, the writer had better cut their teeth on team documentation which starts with small scope and reflect the required information by the stakeholders on the document little by little.

It is clearly expected for the writer to redo the work from the scratch if they try to deliver a magnificent document at once without a blueprint and milestones.

Instead, they should settle down to start it with a small imaginable deliverables and grow them over a long period of time.

2. Top 5 Inevitable Trade-Offs

4. Responsibility of Continuous Maintenance

Once the writer start corporate technical documentation, the quality of the document must stay greater than a certain level to provide benefits; otherwise, it is worthless.

Obsolete documents make confusions after falling behind the latest requirements, which is nothing but a harmful rubbish.

Leaving the rubbish be does such harm as maintenance costs and noises in searching.

In the worst case, the costs invested in the documentation make no sense and the writer could be forced to make another corporate technical document from the scratch, which will definitely lead to additional costs.

That is why it is highly important to keep the standard quality of the document.

The chances are that most of organisations are consciously or unconsciously aware of the fact mentioned above so they are not willing to work on it.

2. Top 5 Inevitable Trade-Offs

4. Responsibility of Continuous Maintenance

The best solution to the trade-off is to choose both of the following strategies.

1. **To make sure of quality of a corporate technical document and the allocated costs with someone superior and authorised such as the writer's boss or manager.**
 - [Top 6 Required Factors for Corporate Technical Documentation](#) must be secured at least.
2. **Involve other team members and build a coalition to make the work done within the estimated time and costs.**
 - Be aware of the shortest path to delivery of the required quality.
3. **Once management costs come not to pay, the writer is highly recommended to archive the document and what values they would like to enjoy by put the requirements, knowledge and management processes into words.**

2. Top 5 Inevitable Trade-Offs

5. Difficulty in Find Out the Upper Limit of Investment

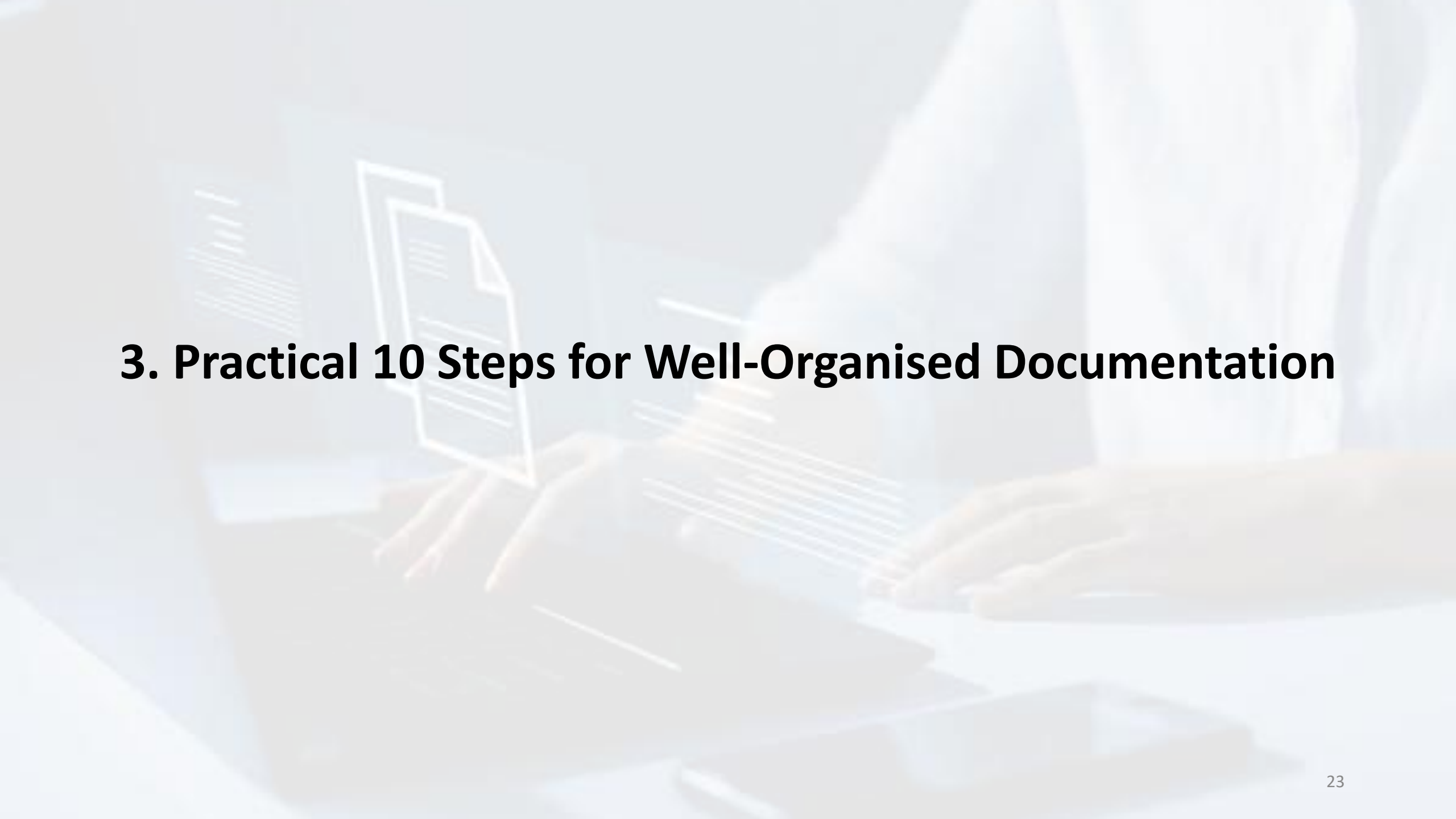
The fact that corporate technical documentation itself earns no money make it difficult to accurately judge ROI(Return of Investment).

There are often gaps between the budget and the actual costs due to potential contingencies which turns out after the documentation gets started.

That is the very reason the work strongly requires the write to have [an ability to manage resources and costs](#).

Unfortunately, there is no silver bullet to this trade-off and all the writer is to undergo a lot of corporate technical documentation activities not just personally but as a team.

Detection of contingencies requires them not just acquisition of theories but practices.

A blurred background image showing a person's hands typing on a laptop keyboard. A white outline of a document with lines of text is overlaid on the left side of the image.

3. Practical 10 Steps for Well-Organised Documentation

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Considering [Top 5 Inevitable Trade-Offs](#) in the last chapter, the writer is supposed to take the following 10 practical steps to maximise the benefits from a corporate technical document.

1. [To Clarify 5Ws and 1H](#)
2. [To Clarify Ownership and Stakeholders](#)
3. [To Determine Where the Document is Placed and Published](#)
4. [To List Up Information to Be on Document](#)
5. [To Design Overall Structure](#)
6. [To Agree with What Format and Description Tools to Use](#)
7. [To Evaluate Difficulty in Documentation in Each Section and Chapter](#)
8. [To Assign Members to Each Section and Chapter](#)
9. [To Review the Document](#)
10. [To Publish the Document](#)

3. Practical 10 Steps for Well-Organised Documentation

1. To Clarify 5Ws and 1H

The following factors are expected to be clear in any job, not just corporate technical documentation.

- **WHY: Purpose.** Problems to be solved and/or Benefits to be provided
- **WHAT: Tasks.** Things to do in order to achieve WHY
- **WHO: Assignee.** A member or a team responsible for the tasks
- **WHEN: Duration.** Timing to start and finish each task and the entire project
- **WHERE: Host/Store.** A place to save the document and/or open its contents
- **HOW: Procedures.** The way to make progress and things done

3. Practical 10 Steps for Well-Organised Documentation

1. To Clarify 5Ws and 1H

Above all, to clarify WHY is most important.

The writer must understand are the stakeholders trouble with the absence of a corporate technical document and what problems will be solved by working on corporate technical documentation in exchange for [the responsibility of continuous maintenance](#).

They can make sense of necessity of the documentation by asking themselves the following 2 questions.

- 1. Would any stakeholder get in trouble** were someone in charge to pass away right away?
- 2. Would someone in charge get in trouble** were any stakeholder to pass away right away?

If the answer of either question is YES, corporate technical documentation is clearly necessary.

(The rest of 4Hs and 1W are supposed to be clear in the following procedures, so please go on to the next page.)

3. Practical 10 Steps for Well-Organised Documentation

2. To Clarify Ownership and Stakeholders

Ownership means who or what team is in charge of design, composition, review and release phases in workflows of documentation.

If it stays obscure, no one or team cares about the responsibility of continuous maintenance, so it must be declared in the earliest phase.

The stakeholders are the readers of the document and information source providers here.

It is necessary to define and comprehend who the readers are and the scope of information provision.

3. Practical 10 Steps for Well-Organised Documentation

2. To Clarify Ownership and Stakeholders

My division assigns the ownership to each corresponding team in the case of team documentation.

In contrast, it is not fixed in annual maintenance projects and the assignee is traditionally supposed to be in charge of the documentation every year.

The fact used to keep bringing about ambiguous ownership and absence of the document, so I take the ownership by myself and released the fixed version(in exchange for loads of pains at [the cynical recursive dependency of work](#)).

If the writer fails to make the ownership crystal, they had better task to their boss or manager and make an action plan together(in the hope that those superior would help them).

3. Practical 10 Steps for Well-Organised Documentation

3. To Determine Where the Document is Placed and Published The first issue the writer commonly faces is "Where should this document be managed and/or released?"

They are required to compare pros with cons of each documentation tool and determine what to adopt after a careful inspection considering the following 2 costs.

- **Reference Costs:** Any blockers in reading the document?
 - Poor searchability, readability and accessibility
 - Poor reference experiences lead to no reference to the document in the near future.
- **Edit Costs:** Any blockers in updating the document?
 - Appropriate document format for management (e.g., HTML, Markdown, WYSIWYG)
 - Leave certain *inconvenience* by keeping [the review phase](#) if the document must be accurate.

3. Practical 10 Steps for Well-Organised Documentation

3. To Determine Where the Document is Placed and Published

My team adopted a rendering engine called [Honkit](#) and the process in which AWS S3 hosts HTML files parsed from Markdown files and publishes static pages.

This method needs the following 3 steps.

1. Create a topic branch from *main* on the corresponding repository edit the document.
2. [Review](#) the document and merge the topic branch to *main*.
3. GitHub Actions automatically deploys static files onto AWS S3 and publish the document, which is triggered by the action of merger to *main* branch.

In terms of reference costs, VPN connection is no more necessary and the document has its index in the tree structure on the side menu, which secures searchability, readability and accessibility.

In terms of edit costs, updated information is always reviewed by other members, which secures accuracy and reliability of the document.

3. Practical 10 Steps for Well-Organised Documentation

4. To List Up Information to Be on Document

To go back to [5Ws and 1H](#) a little bit, corporate technical documentation is needed under a circumstance where the stakeholders have no source of information they require.

Software engineers involved in development of Software as a Service would like to the following information, for example.

- Architecture Chart of Infrastructure
- Data Sequence Chart between Micro Services
- DB chart of Each Table and Column, Entity-Relation Chart
- Outer System Collaboration Chart
- Troubleshooting

3. Practical 10 Steps for Well-Organised Documentation

5. To Design Overall Structure

The writer should go on to sort [the listed-up information to be on the document](#) by the descending order from large/high to small/low in terms of the amount or importance as show below.

1. The Entire Structure
 1. Infrastructure
 2. Communications between Micro Services
2. DB Related Information
 1. Table Structures
 2. Entity-Relation Chart
3. ...

3. Practical 10 Steps for Well-Organised Documentation

5. To Design Overall Structure

The writer would find out there is no ceiling if they kept pursuing the quality of a corporate technical document, so it is wiser and efficient to roughly outline the entire structure of the document and filter truly required information.

As they detail the contents, some components turns out to be inconsistent.

In such a case, updates of the title and the structure of sections and/or chapters are highly possible and allowed.

The writer do not have to follow the structure for good even once fixed.

What is more, **optimisation adjusted between structure and contents is usually observed, so they do not have to worry about changes or updates on the way.**

3. Practical 10 Steps for Well-Organised Documentation

6. To Agree with What Format and Description Tools to Use

The writing format and the chart-drawing tool vary from member to member in the case of team documentation, which is rarely recognised in personal documentation.

As to the writing format, [the casual style](#) should be allowed if [the readers](#) are supposed to be existing members, whereas [the consultative style](#) is appropriate if they are supposed to be new comers and members from another division or corporation.

Regarding the chart-drawing tool, the writer must secure their authority to read and edit even after the assignee leaves.

For example, if images saved on Google Drive are embedded on the document, reference errors are definitely expected to happen once the assignee's account is deleted.

My team adopted [draw.io](#) which highly compatible with Honkit after some plain-text based engines such as [Mermaid](#) and [PlantUML](#) lost candidacies.

Charts are saved as SVG files, referred to by Markdown files and its versions managed on GitHub.

3. Practical 10 Steps for Well-Organised Documentation

7. To Evaluate Difficulty in Documentation in Each Section and Chapter

A corporate technical document has differences in how much each section or chapter is conceptually abstract and they precisely agree with how difficult they are to verbalise.

My team evaluate the difficulty based on the criterion that **the readers would need charts to make sense of the section or chapter** and scored it by the story points show below.

- [MUST] Too complicate and profound to understand without any chart: 3 - 5
- [SHOULD] Understandable with texts but painstaking without charts: 2 - 3
- [MAY] Charts have no effect: 1 - 2

We compared the story points above with out velocity and determined how many tasks we were going to complete in the sprint.

3. Practical 10 Steps for Well-Organised Documentation

8. To Assign Members to Each Section and Chapter

The most efficient way is for the best-informed member is in charge of the specific section or chapter, but the loads will be heavily concentrated on the member.

In contrast, fair distribution of sections or chapters can cause additional investigation and composition costs if the assignee is not familiar with the specific field.

My team the former method at first, but some members had quickly done their tasks and take others, so we finally switched the latter way.

Corporate technical documentation of an unfamiliar field was seriously hard, needless to say; however, a member earnestly communicated with an expert in another team and successfully put the expertise into words on the document.

The member also enjoyed acquiring new knowledge and overcoming the hardship, so the story suggests that **the writer do not have to be afraid of the documentation of something unknown because the experience can bring some positive feedback.**

3. Practical 10 Steps for Well-Organised Documentation

9. To Review the Document

According to [workflows of corporate technical documentation](#), this phase is in charge of review of the document from the point of view of the readers in terms of its design and contents.

Concretely, the reviewer is expected to give feedback to the writer about the following 4 terms.

- [Is it scientific?](#)
- [Is it efficient?](#)
- [Is it impressive?](#)
- [Are the target readers clear?](#)

3. Practical 10 Steps for Well-Organised Documentation

10. To Publish the Document

After the [review](#) is passed, a corporate technical document proceeds to the release phase and is published based on [workflows of corporate technical documentation](#).

The published document is intended for the stakeholders to acquire required knowledge, so it is worthless unless they do not refer.

After the release, the writer must try their best to inform them of where the document is available and get their feedback in order to enhance the document.

As long as the writer or the team of the ownership properly manage the document, the organisation is sure to enjoy [Top 3 Benefits of Well-Organised Corporate Technical Documentation](#).

A person wearing a white lab coat is shown from the chest down, sitting at a desk and typing on a laptop. The image is faded and has a light blue tint. A white outline of a document with lines of text is overlaid on the left side of the laptop screen. The text '4. Summary' is centered in the middle of the image in a bold, black font.

4. Summary

4. Summary

1. Top 5 Inevitable Trade-Offs

1. Cynical Recursive Dependency of Work
2. Agreements to Make with Team Members
3. The More Stakeholders, The Less Quality
4. Responsibility of Continuous Maintenance
5. Difficulty in Find Out the Upper Limit of Investment

2. Practical 10 Steps for Well-Organised Documentation

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5. Epilogue

5. Epilogue

Corporate technical documentation hardly leads to reasonable assessment regardless its importance and difficulty.

I have an impression that most of software engineers value more coding than documentation as a culture and tradition.

Surely, codes usually reflect the latest designs and requirements as long as they are appropriately maintained, so "Read it!" makes sense.

However, as mentioned in [*To Communicate between Developers and Non-Developer*](#), most tasks cannot be completed only by developers and they are required to be able to put technical expertise into understandable words for non-developers as well.

Fortunately, technical documentation is rooted on my division, some members are good at it and they are comparatively properly evaluated.

5. Epilogue

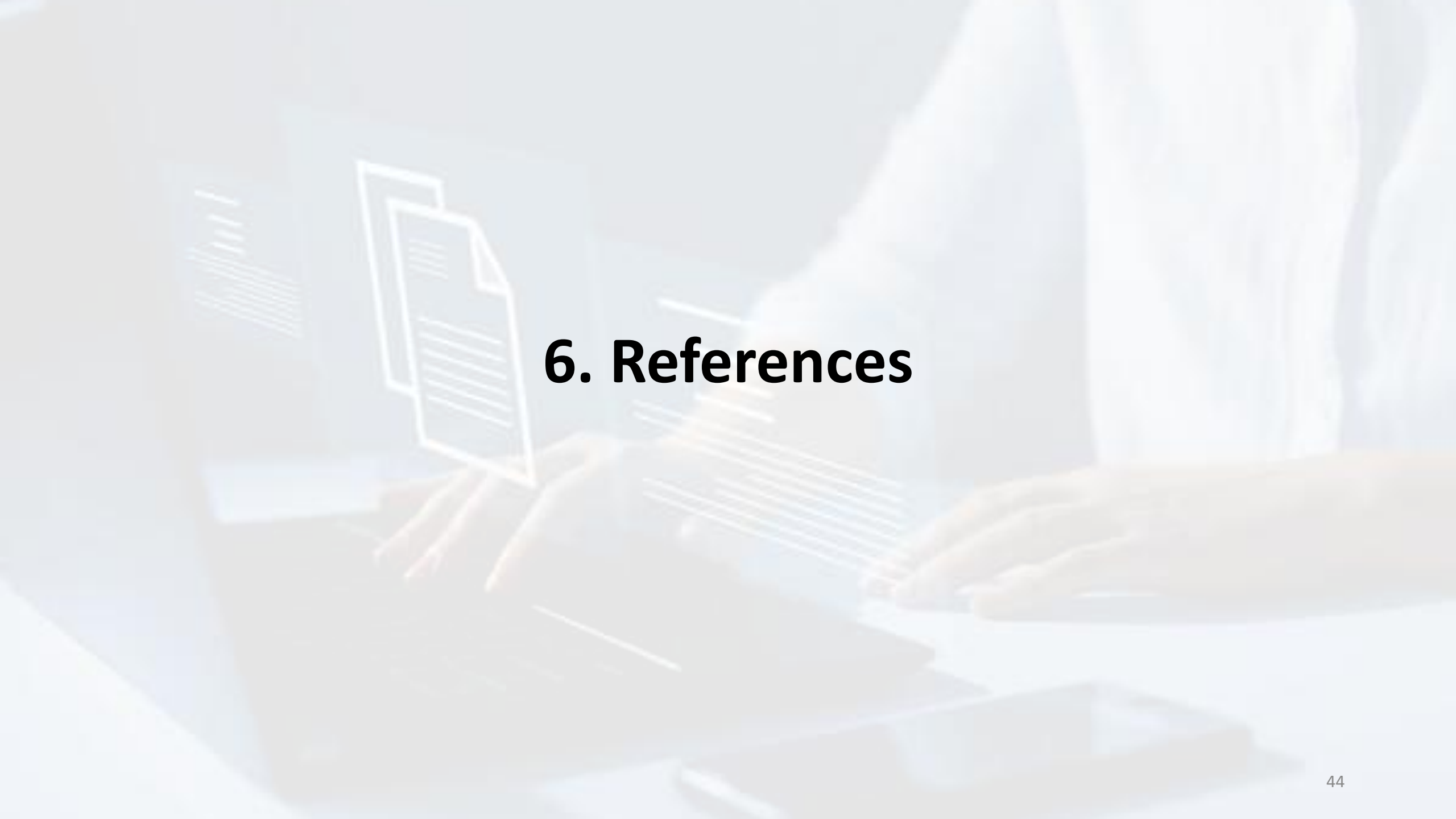
I have worked at 4 companies as of February 2024 and developers who habitually work on documentation has the same tendency **not to trust their own memories**. Source codes you write will be like someone else's ones in a month, so I can assure you that you will hardly remember what contexts you reflected on the codes as an implementor or what made you approve the codes as a reviewer.

Worklogs play an important role in a past-memory and you will be highly thankful for leaving the record to yourself in the past.

Documentation requires a certain level of skills, but it actually concludes **how much you can be kind and gentle to yourself in the future and someone else at present by leaving working logs**.

I suggest that software engineers not good at documentation should cut their teeth on daily working log.

The practice would help you to smoothly work on a corporate technical documentation which is complicated, profound and painstaking.

A person wearing a white lab coat is seated at a desk, typing on a laptop. The laptop screen displays a stylized icon of a document with lines of text. The background is a soft, out-of-focus light blue. The text "6. References" is centered over the image.

6. References

6. References

- [Basics Of Technical Documentation For Engineers](#)
 - Last Accessed On: 12 November 2023
- [Dictionary.com](#)
 - Last Accessed On: 12 November 2023
- [英辞郎Pro Lite](#)
 - Last Accessed On: 12 November 2023
- [Dive Deeper into Present-Day English Vol.5 -Hidden Feature of Language-](#)
 - Last Accessed On: 25 February 2024
- [\[PBI\] 技術仕様ドキュメント作成](#)
 - Accessible Only by Corporate Members
- [\[SBI\]\[技術仕様ドキュメント\] 7. 社外システム連携](#)
 - Accessible Only by Corporate Members

A person wearing a white lab coat is shown from the chest down, sitting at a desk and typing on a laptop. The image is heavily blurred and has a light, ethereal quality. Overlaid on the scene are several white, glowing digital elements: a folder icon with a document inside, a series of horizontal lines, and a stylized representation of a document or data flow. The overall aesthetic is futuristic and technological.

EOD